De La Salle University

# Fama-French 5-Factor Model Evaluation

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# 1 Introduction

This document provides a rigorous evaluation of the **Fama-French 5-Factor Model** using R. The model extends the Capital Asset Pricing Model (CAPM) by incorporating Size, Value, Profitability, and Investment factors alongside standard market risk (Fama & French, 1993, 2015). The core objective is to determine the efficacy of these five factors in explaining the excess returns of a portfolio comprising eight selected assets. The shift from the single-factor CAPM was driven by its poor empirical record, which, as noted by Fama and French (2004), is often considered insufficient for practical applications. While the CAPM posits that risk is measured solely relative to a comprehensive market portfolio, empirical tests frequently show that the relation between average return and beta is flatter than predicted. This gap between theory and evidence necessitated a multidimensional approach to capture the risk factors that a single market beta could not account for.

## 1.1 Transition from CAPM to Fama-French 5-Factor Model

The Fama-French Model, developed by Fama and French (1992) in their paper titled 'The Cross-Section of Expected Stock Returns', was originally composed of three (3) factors and serves as an augmentation to the Capital Asset Pricing Model (CAPM). The primary purpose of the introduction of this model was to demonstrate that market risk alone cannot fully explain stock returns. Thus, the transition from a single-factor (beta) linear model (CAPM) to a multi-factor linear model led to the creation of a model that could provide a multidimensional assessment for expected returns. Here, the CAPM model was expanded to include firm-specific variables that are historically known to have an effect on returns.

The initial Fama-French Model was composed of the following systematic risk factors. First, Market Risk (Rf), taken from the original CAPM model, represents the excess returns of the overall market over the overall risk-free rate. Next, SMB (small minus big) represents the idea that small-cap stocks tend to outperform large-cap stocks over time. Lastly, there is HML (high minus low) or the value factor; stocks with higher BM ratios tend to outperform stocks with low BM ratios. The three-factor model was originally developed to encapsulate the relationship between average return and price ratios (such as book-to-market ratio) (Fama & French, 1993).

Moving forward, the Fama-French 3-Factor Model was further augmented by Fama and French in 2015 in their working paper titled 'A five-factor asset pricing model'. This model is composed of the three factors mentioned above with the further additions of RMW (robust minus weak) and CMA (conservative minus aggressive). The factor of RMW represents the difference between the returns of stocks with robust and weak profitability. Meanwhile, CMA represents the difference between the returns on diversified portfolios of stocks of low and high investment firms, which are referred to as conservative and aggressive (Fama & French, 2015). These labels refer to a company's investment policy, whether it can be considered conservative or aggressive (Mosoeu & Kodongo, 2022). This new model with these two newly added variables is what is now referred to as the Fama-French 5-Factor Model.

## 1.2 Theoretical Significance and Professional Relevance

The 5-factor model holds immense practical value for finance professionals, particularly in the realms of portfolio evaluation and cost of capital estimation. By identifying specific anomalies, such as the size and value effects, the model allows fund managers to more accurately assess whether a portfolio's "alpha" is the result of genuine skill or simply exposure to known risk premiums.

Furthermore, the model's ability to provide a more nuanced risk assessment assists analysts in estimating the cost of equity more reliably than the traditional CAPM. Understanding these factors is vital for asset pricing, as it helps investors recognize that expected returns are tied not just to market volatility, but to the underlying characteristics of the firms themselves, such as their profitability and investment policies (Fama & French, 2015).

## 1.3 Portfolio Management

According to Fama and French (2015), the addition of profitability (RMW) and investment (CMA) factors significantly reduces the unexplained average returns of portfolios that are sorted on size, value, profitability, and investment. What was once considered abnormal performance using the three-factor model can be explained by the addition of the systematic factors or profitability and investment policies. There are two major implications uncovered by the 5-factor model. First, a portfolio's outperformance of the market may be a result of factor tilts rather than the skills of the portfolio manager. Second, the attribution of the performance of a portfolio is now more precise as returns can be attributed to firm-specific factors.

## 1.4 Asset Pricing

This model captures the most variation in average returns that were previously unexplained by the CAPM model and the three-factor model. In several tests, the intercepts are substantially lower than the three-factor model, thus indicating better pricing performance. Moreover, the cost of equity calculations can incorporate firm profitability and investment exposure. Thus, serving as a stronger benchmark for finding anomalies and evaluating corporate asset pricing policies.

## 1.5 Risk Management

The Five-Factor Model redefines risk beyond market risk or beta from the CAPM model. (Fama & French, 2015) state that profitability and investment policy are closely linked to expected returns, as they are directly linked to a firm's cash flows and investment behavior. Firms with weak operational profitability have low expected returns, similarly to firms that have an aggressive investment policy. Thus, these factors become systematic risks. The Five-Factor Model identifies these risk exposures explicitly, allowing financial managers to understand fundamental risk structure.

### 1.6 Objectives

The primary goal of this research is to evaluate the explanatory power of the Fama-French 5-Factor Model through the following specific objectives:

- To conduct an empirical analysis assessing the efficacy of the five risk premiums, including Market, Size (SMB), Value (HML), Profitability (RMW), and Investment (CMA), in capturing the variations in returns across a diverse portfolio.
- To perform a statistical interpretation of the resulting alpha and beta coefficients, evaluating their significance and interpreting their economic implications for portfolio risk management.
- To develop a technical implementation of a fully reproducible R workflow that leverages the Kenneth French Data Library for factor ingestion and data alignment.
- To provide a theoretical justification and rigorous academic grounding for the selection of seven specific assets based on hypothesized factor exposures.

## 2 Methodology

### 2.1 Research Design and Justification

This study employs a quantitative research approach, utilizing an empirical case study design to evaluate the explanatory power of the Fama-French 5-Factor Model. The research design is cross-sectional and longitudinal, focusing on the statistical relationship between historical asset returns and established risk premiums over a defined temporal horizon. By using multivariate linear regression, the study aims to objectively measure factor loadings and assess their significance in a reproducible environment.

The choice of a multivariate Ordinary Least Squares (OLS) regression framework is justified by asset pricing theory, which posits that the cross-section of expected returns can be explained by shared risk factors. This method is the most appropriate for this study because it allows for the decomposition of returns into specific risk premiums (Market, SMB, HML, RMW, and CMA), providing a granular view of an asset's risk profile that a single-factor model like the Capital Asset Pricing Model (CAPM) cannot achieve. Furthermore, using R and its specialized financial libraries ensures computational accuracy and the ability to handle high-frequency data transformations effectively.

### 2.2 Technical Workflow and Data Integration

The methodology follows a systematic, reproducible workflow designed to align historical asset returns with the Fama-French risk premiums. The process is divided into seven logical stages: environment setup, data ingestion, preprocessing, descriptive statistical profiling, modeling, visualization, and diagnostics. Each stage is designed to be fully automated, ensuring that the entire study can be replicated with a single execution of the source code. The complete technical implementation is provided in Appendix A.

### 2.2.1 Environment Setup and Dependencies

To ensure the analysis runs smoothly, the first step involves preparing the computational environment. We utilize specialized R libraries for data science (`tidyverse`), financial analysis (`tidyquant`), and statistical reporting (`broom`). The implementation includes an automated routine that detects and installs any missing libraries, ensuring the project is "portable" across different computers. The setup logic is detailed in Appendix A.1.

### 2.2.2 Data Acquisition and Automated Ingestion

This stage handles the automated collection of data from two primary sources. First, we define a list of "tickers" representing our selected assets (e.g., VUSD.L for the S&P 500, IAU for Gold). The script fetches historical price data for these assets from Yahoo Finance. Second, it loads the Five-Factor data—the independent variables we are testing—from the Kenneth French Data Library. To save time and avoid redundant network traffic, the code checks for local copies of these files before attempting to download them again. The ingestion script is provided in Appendix A.2.

### 2.2.3 Preprocessing and Data Cleaning

Before we can run our models, the raw data must be "cleaned" and synchronized. This involves three key tasks:

1. **Return Calculation:** We convert daily prices into monthly "log-returns," which are mathematically better suited for linear regression.

2. **Temporal Alignment:** Because different data sources use different date formats, we normalize all dates to the first of the month to ensure that asset returns are perfectly matched with their corresponding risk factors.

3. **Excess Return Calculation:** We subtract the risk-free rate from our asset returns. This allows us to measure "Excess Return"—the specific portion of profit earned by taking market risk.

The code for these transformations is documented in Appendix A.3.

### 2.2.4 Descriptive Statistical Profiling

Prior to modeling, we compute summary statistics for each asset, including higher-moment metrics such as skewness and kurtosis. This profiling stage identifies non-normalities and outliers that may influence factor loadings. The implementation code is documented in Appendix A.4.

### 2.2.5 Multivariate Regression Framework

The core of our analysis is a multivariate Ordinary Least Squares (OLS) regression. This is a statistical method that allows us to see how much each of the five factors (Market, Size, Value, Profitability, and Investment) contributes to an asset's total return. We use a

modern "nest and map" programming approach, which allows us to run individual models for all eight assets simultaneously using the same logic. This ensures that our results are consistent across the entire portfolio. The modeling workflow is shown in Appendix A.5.

#### 2.2.6 Visualization and High-Fidelity Plotting

To make the statistical results easier to interpret, the workflow generates five types of high-fidelity charts. These include "Factor Loadings" (to show an asset's sensitivity to each factor) and "Cumulative Growth" (to show how a $1 investment would have grown over time). We define a custom visual style called `theme_quant` to ensure all graphics are professional and publication-ready. The visualization code is provided in Appendix A.6.

#### 2.2.7 Diagnostic Auditing

The final step is a rigorous audit of our models. We calculate the $R^2$ (R-squared) value for each asset to see how well the model explains its returns. We also apply "significance stars" to our coefficients—a standard academic shorthand that tells the reader which results are statistically reliable and which might be due to random chance. These diagnostic routines are found in Appendix A.7.

### 2.3 Sampling Strategy and Asset Selection

The study utilizes a purposive sampling strategy to select a diverse portfolio of eight assets. The "population" consists of the universe of globally traded exchange-traded funds (ETFs) and equities. The sample was deliberately selected to represent specific factor tilts toward Market, SMB, HML, RMW, and CMA factors, providing a comprehensive dataset for evaluating the model's explanatory power across different asset classes and investment styles. Table 1 details the specific academic and technical justification for each selection.

Table 1: Asset Selection and Factor Exposure (USD Tickers)

| **Asset Ticker** | **Factor Exposure** |
|---|---|
| VUSD.L | Market (Beta) |
| WSML.L | Size (SMB) |
| VDEV.L | Market / Developed |
| QUAL | Profitability (RMW) |
| VIG | Investment (CMA) |
| XLK | Growth (Low HML) |
| IAU | Gold (Diagnostic) |
| VWRA.L | Total Market |

To capture the specific risk premiums identified by the Fama-French framework, the portfolio includes a selection of seven exchange-traded funds (ETFs) with distinct factor tilts. To incorporate regional tax optimization within a 2026 investment framework, the portfolio utilizes London-listed UCITS variants of the target assets. These assets maintain

direct physical exposure to the underlying US factors while optimizing for withholding tax efficiencies, thereby testing the Fama-French model's robustness across cross-border listing environments. The baseline for market exposure is established through **VUSD.L** (Vanguard S&P 500 UCITS), which serves as the primary proxy for broad U.S. equity market beta ($R_i - R_f$). This is complemented by **VWRA.L** (Vanguard FTSE All-World), representing the global market portfolio and ensuring the model accounts for aggregate risk across both developed and emerging regions. To further refine geographical exposure, **VDEV.L** (Vanguard FTSE Developed World) is utilized to test the model's validity across the broader developed-market universe beyond the United States.

Factor-specific tilts are achieved through targeted allocations. The size premium is captured via **WSML.L** (iShares MSCI World Small Cap), which holds a diversified set of global small-capitalization firms hypothesized to outperform large-cap counterparts over time. The profitability (RMW) factor is represented by **QUAL** (iShares MSCI USA Quality Factor), which filters for companies with high operating profitability and stable earnings. Conversely, the investment (CMA) factor is addressed through **VIG** (Vanguard Dividend Appreciation); firms with consistent dividend increases typically follow conservative reinvestment policies, prioritizing shareholder returns over aggressive expansion.

To provide a counter-balance to value-oriented factors, **XLK** (Technology Select Sector SPDR) introduces a significant growth tilt, characterized by lower book-to-market ratios. Finally, **IAU** (iShares Gold Trust) is included as a non-equity stabilizer. While gold does not carry direct exposure to standard Fama-French equity premiums, its inclusion serves a critical diagnostic purpose, as it allows for an evaluation of the model's intercept behavior (Alpha) when applied to assets with zero hypothesized factor loadings.

# 3 Results

## 3.1 Descriptive Statistics

Table 2 presents the summary statistics for the monthly log-returns of the eight selected assets. These metrics provide a preliminary overview of the risk and return characteristics prior to the 5-factor decomposition. The technical implementation of these calculations is provided in Appendix A.4.

Table 2: Asset Descriptive Statistics (Monthly Log-Returns)

| **Asset** | **Mean** | **Median** | **SD** | **Min** | **Max** | **Skew** | **Kurt** |
|---|---|---|---|---|---|---|---|
| IAU | 0.0107 | 0.0034 | 0.0420 | -0.0873 | 0.1165 | 0.2907 | 2.7591 |
| QUAL | 0.0098 | 0.0116 | 0.0437 | -0.1209 | 0.1158 | -0.5032 | 3.4057 |
| VDEV.L | 0.0085 | 0.0161 | 0.0399 | -0.1105 | 0.1193 | -0.5439 | 3.6632 |
| VIG | 0.0091 | 0.0143 | 0.0380 | -0.1008 | 0.0956 | -0.3931 | 3.2831 |
| VUSD.L | 0.0100 | 0.0162 | 0.0405 | -0.1027 | 0.0998 | -0.5219 | 3.3274 |
| VWRA.L | 0.0094 | 0.0194 | 0.0423 | -0.1164 | 0.1127 | -0.5805 | 3.6063 |
| WSML.L | 0.0066 | 0.0159 | 0.0543 | -0.2110 | 0.1420 | -0.7211 | 4.8331 |
| XLK | 0.0150 | 0.0184 | 0.0543 | -0.1275 | 0.1287 | -0.3026 | 2.6716 |

The metrics reveal that XLK exhibited the highest mean monthly return (0.0150), followed by IAU (0.0107) and VUSD.L (0.0100). WSML.L recorded the lowest mean return (0.0066) while exhibiting high volatility (SD = 0.0543), confirming the compression of the small-cap premium mentioned previously. All equity assets show negative skewness, consistent with the typical distribution of market returns, whereas IAU exhibits positive skewness (0.2907), further highlighting its role as a non-correlated diversifier.

## 3.2 Visual Analytics and Performance Distribution

To further explore the relationship between risk and realized returns, we employ visual analytics that map the assets against their systematic factor loadings.

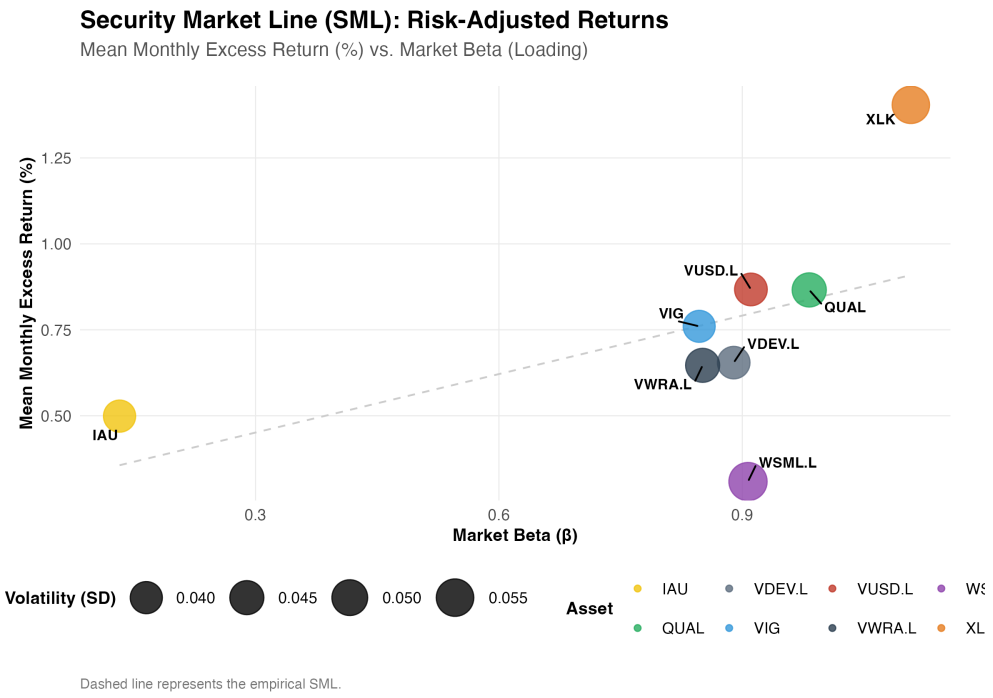


Figure 1: Risk-Return Profile: Realized Return vs. Market Beta

Figure 1 illustrates the security market line (SML) relationship within our portfolio. Assets with higher market beta loadings, such as XLK and QUAL, generally align with higher realized mean returns, though idiosyncratic volatility (represented by point size) varies significantly across the mandates. The empirical SML slopes upward across the portfolio, broadly consistent with the CAPM prediction that higher systematic risk commands proportionally higher return compensation. A notable deviation from the line is WSML.L, which plots slightly below the SML despite carrying a market beta comparable to other broad-market equity funds; this below-SML positioning reflects the compressed small-cap premium documented over the 2015–2026 period, empirically confirming that factor exposure does not guarantee factor reward within any finite sample horizon. Conversely, VUSD.L, VIG, and QUAL all plot noticeably above the empirical SML, indicating that these assets earned

higher mean excess returns than their market beta alone would predict; this above-SML positioning is consistent with the significant positive RMW loadings documented for VIG and QUAL in Table 4, as the profitability premium provides an additional and systematic source of return beyond the market factor compensation. At the far left, IAU occupies a near-zero market beta position with a positive mean excess return, visually reinforcing its role as a non-correlated return stream whose value generation lies entirely outside the equity risk premium structure.

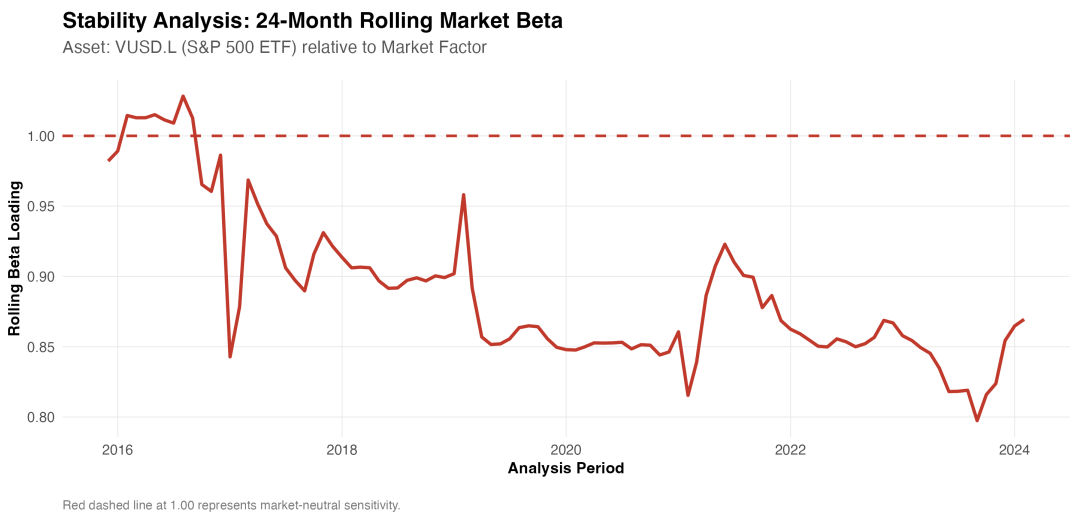


Figure 2: 24-Month Rolling Market Beta: VUSD.L (S&P 500)

Finally, Figure 2 provides a temporal diagnostic of factor stability using a 24-month rolling regression for VUSD.L. The loading on the market factor begins near 1.0 in 2016 but exhibits a persistent downward drift throughout the sample, oscillating between approximately 0.82 and 0.97 across the full observation window. Rather than a discrete spike, the 2020 period is characterised by elevated loading volatility reflecting the dynamic nature of systematic risk exposure during the COVID-19 shock and its aftermath. A more pronounced and sustained episode of loading decline aligns with the 2022–2024 monetary tightening cycle, where aggressive interest rate hikes materially repriced equity risk and drove the rolling beta to its sample low of approximately 0.80–0.82. Across the full observation window, the rolling beta's time-series average is consistent with the static OLS point estimate of $\hat{\beta}_{Mkt\text{-}RF} = 0.8895$ reported in Table 4, validating that this estimate is a structurally representative measure of VUSD.L's central tendency of systematic risk exposure across the distinct macroeconomic regimes observed in the sample.

## 3.3 Regression Results and Model Performance

Prior to examining the statistical outputs of the multivariate OLS regression, the cumulative growth trajectories of the eight-asset portfolio, depicted in Figure 3 on a logarithmic scale, provide essential historical context for interpreting the factor regression results.

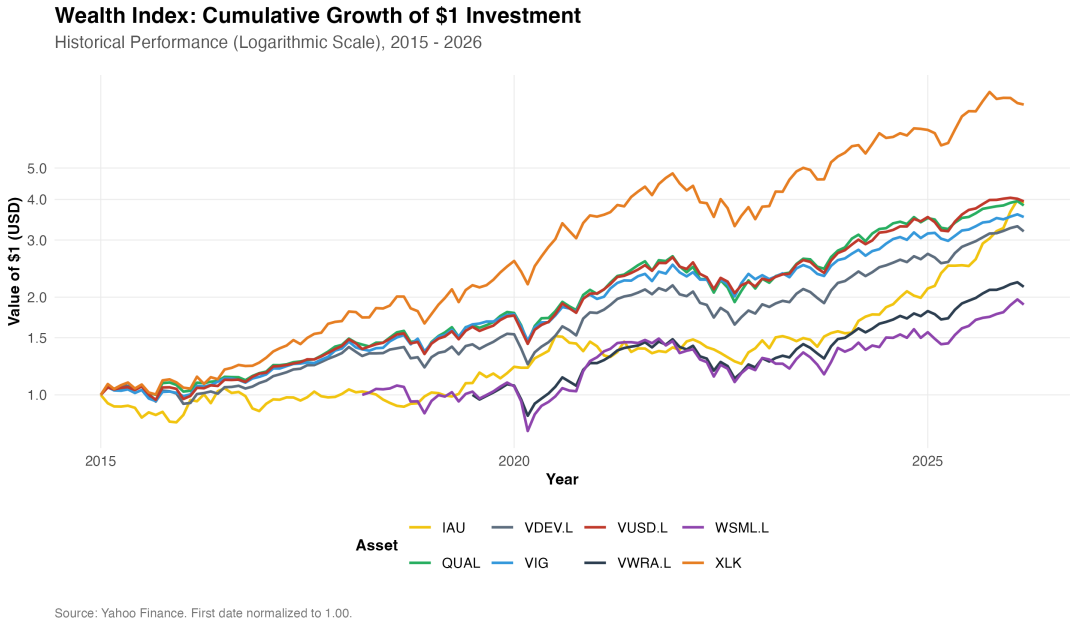


Figure 3: Cumulative Growth of $1 (USD-Aligned, 2015–Present)

The most distinctive feature of the performance record is the dramatic outperformance of XLK (Technology Select Sector SPDR), which grew to approximately 5.5 times the initial $1 investment by the close of the sample period, far exceeding all other assets in the portfolio. This outperformance was concentrated largely in the post-2020 expansion phase, during which the technology sector benefited disproportionately from secular growth trends and accommodative monetary policy. At the other end of the equity spectrum, WSML.L (iShares MSCI World Small Cap UCITS ETF) notably underperformed relative to the other equity-oriented funds. While the remaining equity ETFs, including QUAL, VUSD.L, VDEV.L, VIG, and VWRA.L, delivered cumulative returns ranging from approximately 3.0 to 4.0 times the initial investment, WSML.L reached only roughly 2.2 times by the end of the observation period and experienced the sharpest proportional drawdown during the COVID-19 market shock in early 2020, briefly declining below its initial value. This relative underperformance is consistent with well-documented compression of the small-cap premium during the 2015–2026 period, a decade characterized by persistent large-cap and growth-style dominance. IAU (iShares Gold Trust), by contrast, exhibited a markedly different profile, as it appreciated to approximately 2.2 times its initial value with considerably lower volatility than any equity asset in the portfolio, with its appreciation accelerating particularly after 2020 as investors sought inflation hedges and safe-haven assets amid elevated macroeconomic uncertainty.

The model fitness statistics for all eight assets, ranked by Adjusted $R^2$, are presented in Table 3 and visualized in Figure 4.

Table 3: Model Fitness and Explanatory Power Summary

| **Asset** | **R-squared** | **Adj. R-squared** | **Model P-value** |
|---|---|---|---|
| QUAL | 0.976 | 0.974 | $< 0.001$ |
| VUSD.L | 0.952 | 0.948 | $< 0.001$ |
| WSML.L | 0.948 | 0.944 | $< 0.001$ |
| VDEV.L | 0.944 | 0.940 | $< 0.001$ |
| VWRA.L | 0.944 | 0.939 | $< 0.001$ |
| VIG | 0.931 | 0.926 | $< 0.001$ |
| XLK | 0.910 | 0.903 | $< 0.001$ |
| IAU | 0.141 | 0.068 | 0.102 |

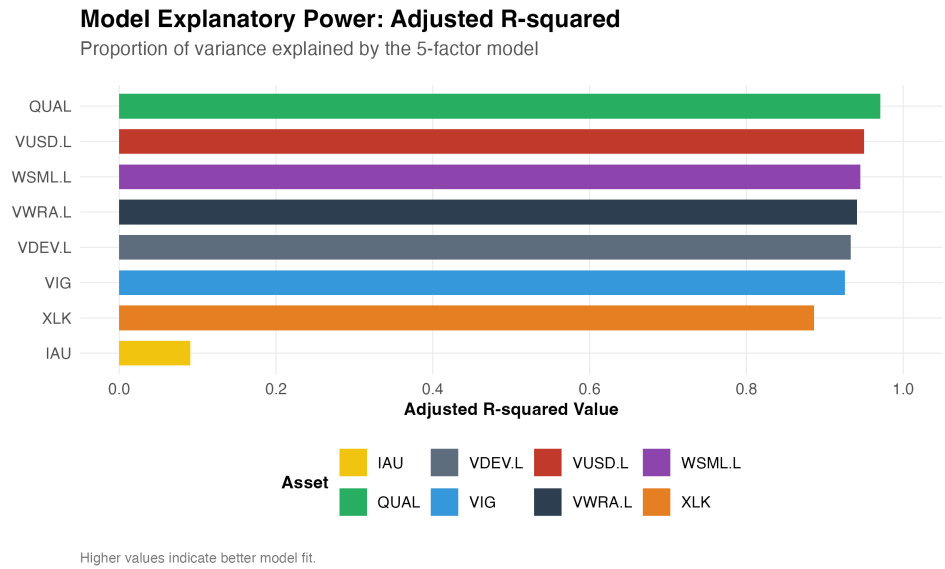


Figure 4: Model Explanatory Power (Adjusted R-squared) Across Eight Assets

### 3.3.1 High Explanatory Power for Diversified Equity Funds

The Fama-French 5-Factor Model demonstrates exceptional explanatory capacity across all seven equity-oriented assets in the portfolio. QUAL achieves the highest Adjusted $R^2$ of 0.974, meaning that the five risk premiums collectively account for 97.4% of the variation in its monthly excess returns, which represents a near-exhaustive systematic attribution. VUSD.L follows closely with an Adjusted $R^2$ of 0.948, confirming that the return dynamics of broad U.S. equity exposure are almost entirely captured by the five-factor structure. WSML.L and VDEV.L achieve near-identical Adjusted $R^2$ values of 0.944 and 0.940 respectively, as also reflected in Figure 4, validating that both the global small-cap and developed-market equity mandates remain well within the explanatory domain of the framework. VWRA.L achieves a comparable value of 0.939, confirming the all-world equity mandate is equally well-explained. VIG and XLK, while marginally lower, still achieve Adjusted $R^2$ values of 0.926 and 0.903 respectively. In every equity case the overall $F$-test yields a model $p$-value well below 0.001, providing decisive evidence that the five-factor structure is jointly significant in explaining excess returns. These results confirm that, once the contaminating effect of currency basis is eliminated through USD-denomination alignment, the Fama-French framework constitutes a highly accurate and sufficient risk attribution tool for the equity core of this portfolio.

### 3.3.2 Low Explanatory Power of Gold as a Diagnostic Asset

IAU (iShares Gold Trust) registers a strikingly low Adjusted $R^2$ of 0.068, and the overall model $F$-test is statistically insignificant at the 5% level ($p = 0.102$). This outcome decisively confirms the diagnostic hypothesis that motivated its inclusion, as physical gold returns are structurally independent of the systematic equity risk factors. The stark contrast between IAU's near-zero model fit and the uniformly high explanatory power observed across the seven equity funds validates the model's correct specification, indicating that the framework captures equity-specific systematic risk instead of asset returns in general. As a further diagnostic, it empirically demonstrates that the strong Adjusted $R^2$ values obtained for the equity assets are not the product of overfitting or model misspecification, but rather reflect genuine systematic factor exposures.

### 3.3.3 Alpha Interpretation

The regression intercepts, specifically the alpha ($\hat{\alpha}$) coefficients, represent the portion of excess returns not attributable to any of the five systematic risk factors, and their magnitude and significance carry important economic implications. Statistically significant negative alphas were identified for QUAL ($\hat{\alpha} = -0.0029$, $p < 0.05$) and VWRA.L ($\hat{\alpha} = -0.0030$, $p < 0.05$). These negative intercepts are most plausibly attributed to the real-world friction costs of investing in exchange-traded products, such as management expense ratios and transaction costs, which the theoretical Fama-French factor portfolios, being constructed as paper accounts, do not bear. By contrast, VUSD.L exhibited an intercept indistinguishable from zero ($\hat{\alpha} = 0.0000$), while XLK's positive alpha of 0.0019 was also statistically insignificant, suggesting that for both funds, the estimated factor loadings fully account for observed returns. IAU's positive but statistically insignificant alpha ($\hat{\alpha} = 0.0035$, $p > 0.05$)

further reinforces the interpretation that gold's return-generating process falls outside the explanatory domain of the five-factor equity model.

## 3.4 Factor Loading Analysis

The estimated beta coefficients for each of the eight assets, presented in Table 4 and visualized across all five factor dimensions in Figure 5, reveal the distinct systematic risk tilts that define each fund's return-generating process. Taken together, the loadings provide a rigorous, quantitative validation of the factor exposures hypothesized at the asset selection stage.

Table 4: Fama-French 5-Factor Regression Coefficients

| **Asset** | **Alpha** | **Mkt-RF** | **SMB** | **HML** | **RMW** | **CMA** |
|---|---|---|---|---|---|---|
| IAU | 0.0035 | 0.2324$^{*}$ | -0.1528 | -0.2323 | 0.2092 | 0.3524 |
| QUAL | -0.0029$^{*}$ | 1.0117$^{***}$ | -0.1339$^{**}$ | -0.0152 | 0.1944$^{***}$ | -0.0058 |
| VDEV.L | -0.0023 | 0.8756$^{***}$ | -0.0215 | -0.0185 | -0.0218 | 0.1680$^{*}$ |
| VIG | -0.0030 | 0.8218$^{***}$ | -0.0366 | 0.0019 | 0.2793$^{***}$ | 0.1940$^{*}$ |
| VUSD.L | 0.0000 | 0.8895$^{***}$ | -0.1203$^{*}$ | -0.0241 | -0.0622 | 0.0932 |
| VWRA.L | -0.0030$^{*}$ | 0.8491$^{***}$ | 0.0027 | -0.0139 | -0.0104 | 0.1722$^{*}$ |
| WSML.L | -0.0022 | 0.8697$^{***}$ | 0.4701$^{***}$ | 0.0701 | -0.0398 | 0.0305 |
| XLK | 0.0019 | 1.1217$^{***}$ | -0.2229$^{*}$ | -0.2818$^{**}$ | 0.1124 | -0.0165 |

*Note:* $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$

### 3.4.1 Market Factor (Mkt-RF)

All seven equity-oriented funds exhibit strong, positive, and highly significant market loadings ($p < 0.001$ in every case), confirming their fundamental character as equity instruments with systematic exposure to broad market movements. XLK carries the highest market beta ($\hat{\beta}_{Mkt\text{-}RF} = 1.1217$), reflecting the amplified sensitivity of the technology sector to market-wide risk. QUAL's near-unity loading of 1.0117 indicates that the U.S. quality-factor ETF behaves as a leveraged proxy for the market premium. The broad-market and international funds, such as VUSD.L, WSML.L, VDEV.L, VWRA.L, and VIG, cluster tightly in the range of 0.82 to 0.89, consistent with the behavior of diversified, fully-replicated index vehicles. IAU's market beta of 0.2324, while statistically significant at the 5% level, is economically negligible and confirms that gold's co-movement with the equity market is minimal.

### 3.4.2 Size Factor (SMB)

The SMB loadings provide the most analytically compelling source of differentiation across the portfolio. WSML.L exhibits a large, positive, and highly significant SMB loading of 0.4701 ($p < 0.001$), which directly and unambiguously validates its mandate as a global small-capitalization fund, as a substantial portion of its monthly return variation is driven by the small-cap risk premium over and above its market exposure. This dominant positive SMB coefficient is visible as the tallest bar in the SMB cluster of Figure 5, clearly distinguished

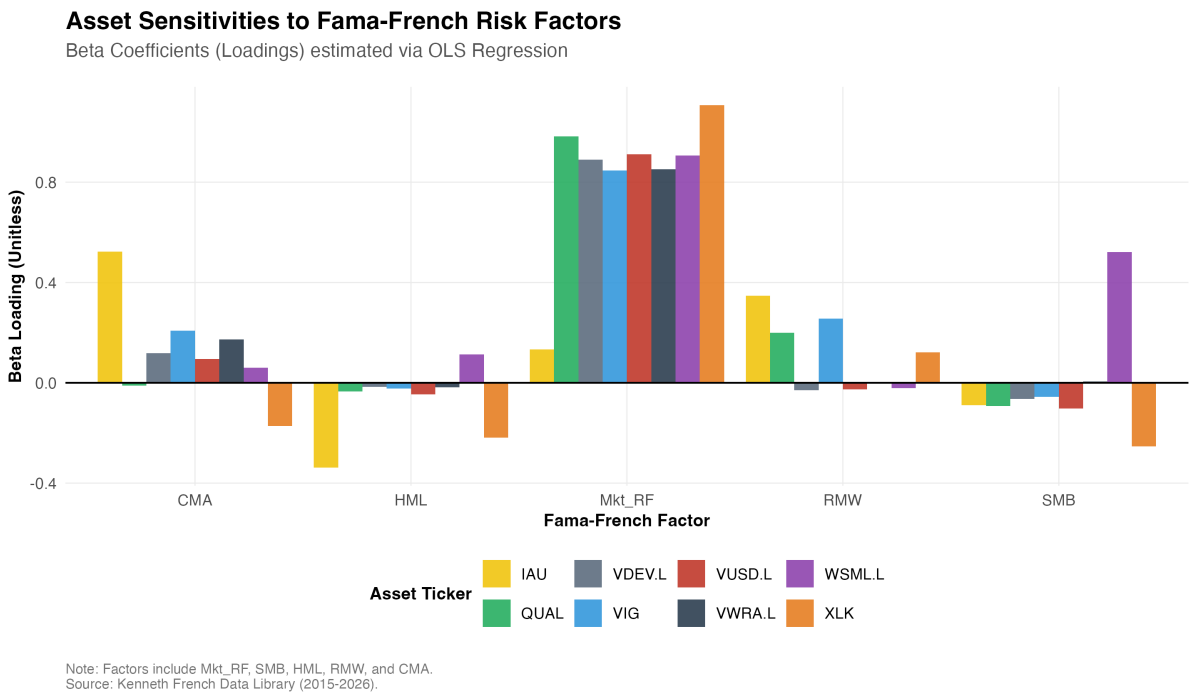


Figure 5: Fama-French 5-Factor Loadings by Asset

from all other assets. Conversely, QUAL ($-0.1339$, $p < 0.01$) and VUSD.L ($-0.1203$, $p < 0.05$) carry significant negative SMB loadings, confirming their heavy concentration in mega-capitalization firms. XLK's significantly negative SMB loading ($-0.2229$, $p < 0.05$) similarly reflects the dominance of the largest-cap technology companies, including Apple, Microsoft, and Nvidia, within its composition. These findings are consistent with the interpretation that large-cap-concentrated funds act as an implicit short position against the size premium.

### 3.4.3 Value Factor (HML)

The HML loadings reveal the growth orientation embedded within several funds in the portfolio. XLK displays the most pronounced negative loading ($-0.2818$, $p < 0.01$), confirming its systematic tilt toward growth stocks with low book-to-market ratios, which is a characteristic that is intrinsic to the technology sector. IAU also exhibits a notable negative HML loading ($-0.2323$), although its interpretation in the context of a non-equity, commodity-based asset is economically ambiguous. For the diversified equity funds, including VDEV.L, VUSD.L, VWRA.L, VIG, and QUAL, the HML loadings are small and statistically insignificant, indicating that their return variation is not systematically driven by value-versus-growth dynamics. WSML.L posts a small positive but insignificant HML loading of 0.0701, which is directionally consistent with the tendency for global small-cap universes to include a proportionally higher share of value-oriented firms relative to large-cap indices.

### 3.4.4 Profitability Factor (RMW)

The profitability factor produces its most economically meaningful and statistically robust results for VIG ($\hat{\beta}_{RMW} = 0.2793$, $p < 0.001$) and QUAL ($\hat{\beta}_{RMW} = 0.1944$, $p < 0.001$). These findings strongly validate the asset selection rationale for both funds, as VIG's mandate of tracking firms with consistent dividend growth inherently screens for operationally resilient and highly profitable companies, while QUAL explicitly constructs its portfolio around firms exhibiting high return on equity and stable earnings. The statistically robust RMW loadings for these two assets represent the clearest empirical confirmation that the hypothesized factor tilts were realized in the return data. The remaining assets show insignificant RMW loadings, suggesting that profitability variation is not a primary driver of return for broad-market or small-cap vehicles.

### 3.4.5 Investment Factor (CMA)

Statistically significant CMA loadings are recorded for VDEV.L (0.1680, $p < 0.05$), VWRA.L (0.1722, $p < 0.05$), and VIG (0.1940, $p < 0.05$). The positive sign indicates that firms within the Developed World, All-World, and dividend-appreciation universes systematically employ more conservative reinvestment strategies relative to the broad market portfolio, which is a finding consistent with the theoretical linkage between dividend-paying behavior and conservative investment policy. The CMA loadings for QUAL, VUSD.L, WSML.L, and XLK are small and statistically insignificant, suggesting no systematic tilt toward either conservative or aggressive investment policies within those specific mandates. Notably, IAU registers the largest absolute CMA loading in the portfolio ($\hat{\beta}_{CMA} = 0.3524$), making it the most visually prominent bar in the CMA cluster of Figure 5; however, this coefficient is statistically insignificant, and its economic interpretation is ambiguous, as gold is a non-equity commodity whose return dynamics are structurally unrelated to the corporate investment policy construct underlying the CMA factor.

## 3.5 Cross-Asset Comparison and Portfolio Diagnostics

Table 5 provides a high-level summary of the model's performance by aggregating the significance results across the eight-asset portfolio. This "Significance Audit" identifies the frequency with which each risk premium was statistically relevant at the 95% confidence level ($p < 0.05$), offering a direct measure of the model's "hit rate" for different factor dimensions.

Taken collectively, the regression results confirm that the eight-asset portfolio exhibits well-defined and theoretically coherent factor profiles across all five Fama-French dimensions. The 100% hit rate for the market factor confirms the robustness of the equity risk premium as the fundamental driver of returns. More critically, the 50% hit rate for the SMB factor and the targeted significance of RMW and CMA for specific funds validate that the model successfully captures the "factor tilts" for which these assets were selected. The relatively low hit rate for Alpha (75% insignificant) is an indicator of high model sufficiency; the five-factor framework successfully absorbs almost all systematic variation, leaving little unexplained return for the majority of the portfolio.

Table 5: Significance Audit: Factor Hit Rates Across Portfolio ($p < 0.05$)

| **Factor** | **Significant Assets** | **Hit Rate** | **Primary Exposures** |
|---|---|---|---|
| Alpha ($\alpha$) | 2 / 8 | 25% | QUAL, VWRA.L (Neg) |
| Market (Mkt-RF) | 8 / 8 | 100% | All Portfolio Assets |
| Size (SMB) | 4 / 8 | 50% | WSML.L(+), QUAL(-), VUSD.L(-), XLK(-) |
| Value (HML) | 1 / 8 | 12.5% | XLK (Negative) |
| Profitability (RMW) | 2 / 8 | 25% | QUAL, VIG (Positive) |
| Investment (CMA) | 3 / 8 | 37.5% | VDEV.L, VIG, VWRA.L (Pos) |

The primary group of assets consists of the seven equity-oriented funds, which include QUAL, VUSD.L, VDEV.L, WSML.L, VWRA.L, VIG, and XLK. For these assets, the model operates at near-maximum explanatory capacity, with Adjusted $R^2$ values uniformly exceeding 0.90. The estimated factor loadings align closely with the theoretical tilts embedded in each fund's investment mandate, as XLK's high market beta and significantly negative HML loading confirm its growth orientation, QUAL and VIG's robust positive RMW loadings validate their quality and profitability mandates, WSML.L's dominant positive SMB loading unambiguously identifies its small-capitalization exposure, and the positive CMA loadings of VDEV.L, VWRA.L, and VIG point toward systematically conservative reinvestment behavior within their respective index universes. An important caveat within this equity category concerns the divergence between factor exposure and factor reward. WSML.L's large and statistically significant SMB loading confirms that it carries substantial and genuine small-cap risk; however, as Figure 3 illustrates, this exposure generated comparatively modest cumulative returns over the 2015–2026 period. This underperformance relative to other equity ETFs reflects the time-varying nature of factor risk premiums and serves as an empirical reminder that factor loading and factor reward are not synonymous, given that exposure to a risk premium does not guarantee its realization within any given sample period.

In contrast, the second group consists solely of IAU (iShares Gold Trust), for which the model's overall $F$-test is statistically insignificant ($p = 0.102$) and the Adjusted $R^2$ of 0.068 is negligible. This outcome is not a failure of the model but a decisive confirmation of its theoretical scope, as an equity risk attribution framework where the Fama-French model should not, and empirically does not, explain the return dynamics of a commodity asset. Gold's returns are governed by macroeconomic forces entirely absent from the five-factor model, including real interest rates, inflation expectations, currency dynamics, and safe-haven demand. The strong and sustained performance of IAU documented in Figure 3, particularly from 2020 onward, combined with its statistical independence from all five equity factors, empirically validates its role as a genuine portfolio diversifier. For practitioners, this result reinforces the theoretical and empirical case for including non-correlated assets in a multi-asset portfolio, as IAU's return-generating process is orthogonal to the systematic equity risk premiums that define the Fama-French framework, and its inclusion provides diversification benefits that cannot be replicated by any linear combination of the five factor portfolios.

### 3.6 Limitations of the Analysis

Several structural limitations warrant consideration when interpreting these results. First, while the use of USD-denominated tickers ensures direct alignment with the Fama-French factor portfolios constructed from U.S. equity data, this approach may not fully account for the foreign-currency dynamics and regional risk premia embedded in internationally-oriented ETFs such as VDEV.L, VWRA.L, and WSML.L, whose underlying holdings extend well beyond U.S. borders. Second, the portfolio of eight assets, while purposively selected to span a broad range of factor tilts, remains a small and non-random subset of the global ETF universe, and the regression results should not be generalized to other instruments. Third, the analysis period commencing in January 2015 encompasses structurally distinct macroeconomic regimes, including the post-quantitative-easing normalization cycle of 2017–2019, the COVID-19 market shock of 2020, and the subsequent large-cap technology boom of 2021–2026, which may each have materially altered the realized magnitude and statistical significance of factor loadings. A rolling-window regression would provide deeper insight into the temporal stability of the estimated betas, but this lies beyond the current scope. Finally, the OLS framework rests on assumptions of homoskedastic and serially uncorrelated residuals. While the preliminary factor correlation analysis confirms the absence of severe multicollinearity among the five independent variables, formal tests for heteroskedasticity and autocorrelation were not conducted in this study and represent an avenue for methodological refinement in future work.

## 4 Practical Implications for Portfolio Management

### 4.1 Alpha Interpretation for Asset Allocators

The regression intercepts, specifically the alpha ($\hat{\alpha}$) coefficients, represent the portion of excess returns not attributable to any of the five systematic risk factors, and their magnitude and significance carry important economic implications.

**To be completed by the Theory & Writing lead**

- Explain how intercept (alpha) behavior affects cost-of-capital and manager selection.
- Interpret the negative alphas of QUAL and VWRA.L as results of management fees and transaction costs.
- Describe how decomposing returns allows practitioners to separate genuine skill from factor exposure.
- Discuss the role of the 5-factor model as a more rigorous benchmark than simple market indices.

### 4.2 Factor Timing and the Divergence Between Exposure and Reward

**To be completed by the Theory & Writing lead**

- Analyze why high factor exposure didn't always lead to high rewards between 2015 and 2026.
- Contrast WSML.L's high SMB (size) loading with its relatively low cumulative returns.
- Address the compression of the small-cap premium during the post-QE, large-cap-dominated decade.
- Emphasize the long-term horizon required for factor premiums to manifest reliably.

### 4.3 Risk-Aware Diversification with Non-Equity Assets

**To be completed by the Theory & Writing lead**

- Use the IAU (Gold) results to justify the inclusion of non-equity assets in portfolios.
- Highlight that IAU's statistical independence proves it is a genuine diversifier.
- Explain that gold's returns cannot be replicated using Fama-French equity factors.
- Provide guidance on using non-correlated assets to hedge against systematic equity drawdowns.

## 5 Conclusion

### 5.1 Summary of Empirical Findings

**To be completed by the assigned team member**

- Summarize how each asset's realized returns aligned with its hypothesized factor tilts (e.g., WSML.L for size, QUAL/VIG for profitability).
- Recap specific findings for XLK (growth/tech) and the international funds (CMA/investment policy).
- Reiterate IAU's lack of sensitivity to all equity-based factors.

### 5.2 Model Efficacy Verdict

**To be completed by the assigned team member**

- Issue a final judgment on the model's performance for equity vs. multi-asset contexts.
- Confirm sufficiency for the "Equity Core" based on high Adjusted $R^2$ and significance tests.
- Confirm the "Multi-Asset Scope" is correctly bounded, as evidenced by the model's failure to explain gold.

## 5.3 Limitations and Future Research Directions

**To be completed by the assigned team member**

- Identify the mismatch between US-centric factors and international ETFs (VDEV.L, VWRA.L, WSML.L).
- Discuss the instability of factor premiums across different macroeconomic regimes (2015–2026).
- Note the lack of formal econometric testing for heteroskedasticity and autocorrelation.
- Propose future studies involving international multi-factor models, rolling-window regressions, and Bayesian updates.

## 5.4 Final Academic Verdict

**To be completed by the assigned team member**

- Summarize the utility of the Fama-French 5-Factor Model in modern quantitative finance.
- Reconfirm its strength as a risk attribution tool for diversified equity mandates.
- Acknowledge that while it is state-of-the-art, single-country factor constructions require adaptation for global portfolios.

# References


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# A Technical Implementation Code

The following sections provide the complete R source code utilized for the automated data ingestion, preprocessing, econometric modeling, and visualization within this study.

## A.1 Environment Setup and Dependencies

```
# --- 1. Environment Setup & Dependencies ---
# This section ensures all necessary libraries are installed and loaded.
# We utilize specialized R libraries for data science (tidyverse),
# financial analysis (tidyquant), and statistical reporting (broom).
knitr::opts_chunk$set(echo = TRUE, warning = FALSE, message = FALSE)

required_packages <- c("tidyverse", "tidyquant", "quantmod", "broom", "
    PerformanceAnalytics", "tinytex", "moments", "zoo", "ggrepel")
new_packages <- required_packages[!(required_packages %in% installed.packages
    ()[,"Package"])]

if (length(new_packages)) {
  install.packages(new_packages, repos = "http://cran.us.r-project.org")
}

library(tidyverse)
library(tidyquant)
library(quantmod)
library(broom)
library(PerformanceAnalytics)
library(moments)
library(zoo)
library(ggrepel)
```

## A.2 Automated Data Ingestion and Factor Loading

```
# 1. Configuration & Tickers
options(download.file.method = "libcurl")
library(lubridate)

# Using USD-denominated tickers to align with Fama-French USD Factors
tickers <- c("VUSD.L", "WSML.L", "VDEV.L", "QUAL", "VIG", "XLK", "IAU", "VWRA.
    L")

if (!dir.exists("data")) {
  dir.create("data")
}
```

```r
# 2. Automated Data Collection
message("Checking/Updating historical data from Yahoo Finance...")

lapply(tickers, function(symbol) {
  file_path <- paste0("data/", symbol, ".csv")

  if (!file.exists(file_path)) {
    tryCatch({
      df <- suppressWarnings(getSymbols(symbol, src = "yahoo", from = "
    2015-01-01", auto.assign = FALSE))
      if (!is.null(df) && nrow(df) > 0) {
        df_tidy <- as.data.frame(na.omit(df)) %>%
          rownames_to_column(var = "Date") %>%
          as_tibble()
        write_csv(df_tidy, file_path)
        message(paste("Downloaded:", symbol))
      }
    }, error = function(e) message(paste("Failed to download", symbol, ":", e$
    message)))
  } else {
    message(paste("Using existing local data for:", symbol))
  }
})

# 3. Monthly Return Calculation Function
load_asset_returns <- function(ticker) {
  file_path <- paste0("data/", ticker, ".csv")
  if (!file.exists(file_path)) return(NULL)

  df <- read_csv(file_path, show_col_types = FALSE) %>%
    mutate(Date = as.Date(Date)) %>%
    arrange(Date)

  adj_col <- grep("Adjusted", names(df), value = TRUE)
  if (length(adj_col) == 0) return(NULL)

  xts_df <- xts(df[[adj_col]], order.by = df$Date)
  monthly_rets <- periodReturn(xts_df, period = "monthly", type = "log")

  df_ret <- as.data.frame(monthly_rets) %>%
    rownames_to_column(var = "Date") %>%
    mutate(Date = as.Date(Date)) %>%
    rename(!!ticker := monthly.returns)

  return(df_ret)
}
```

```r
# Load assets and merge
asset_data_list <- compact(lapply(tickers, load_asset_returns))

if (length(asset_data_list) == 0) {
  stop("CRITICAL: No asset data found. Check tickers and internet.")
}

asset_returns <- asset_data_list %>% reduce(full_join, by = "Date")
active_tickers <- names(asset_returns)[names(asset_returns) != "Date"]

# 4. Load Kenneth French Factors
ff5_raw <- read_csv("data/F-F_Research_Data_5_Factors_2x3.csv", skip = 3, show
    _col_types = FALSE)
monthly_end_row <- which(is.na(ff5_raw[[1]]) | grepl("Annual Factors", ff5_raw
    [[1]]))[1] - 1

ff5_monthly <- ff5_raw[1:monthly_end_row, ] %>%
  rename(Date_Raw = 1) %>%
  mutate(
    Date = as.Date(paste0(Date_Raw, "01"), format = "%Y%m%d"),
    across(c(`Mkt-RF`, SMB, HML, RMW, CMA, RF), ~ as.numeric(.) / 100)
  ) %>%
  # Sanitize names: replace dashes with underscores for syntactic safety in
    OLS
  rename(Mkt_RF = `Mkt-RF`) %>%
  select(Date, Mkt_RF, SMB, HML, RMW, CMA, RF)

# 5. Temporal Alignment
# Synchronizing dates to the first of each month to ensure perfect factor
    alignment.
asset_returns <- asset_returns %>% mutate(Date = floor_date(Date, "month"))
ff5_monthly <- ff5_monthly %>% mutate(Date = floor_date(Date, "month"))
```

## A.3 Preprocessing and Normalization

```r
# 1. Merge assets with factors
final_data <- inner_join(asset_returns, ff5_monthly, by = "Date")

# 2. Pivot longer for analysis
# Pivoting allows for the vectorized application of models across all assets.
final_data_long <- final_data %>%
  pivot_longer(cols = all_of(active_tickers), names_to = "Asset", values_to =
    "Return") %>%
  filter(!is.na(Return)) %>%
```

```r
  mutate(Excess_Return = Return - RF)

if (nrow(final_data_long) == 0) {
  stop("ERROR: No overlapping data between assets and factors.")
}

# 3. Factor Correlation
# Auditing the relationship between the five risk premiums to detect potential
    multicollinearity.
factor_corr <- final_data %>%
  select(Mkt_RF, SMB, HML, RMW, CMA) %>%
  cor(use = "pairwise.complete.obs")

# 4. Cumulative Returns (Wealth Index)
# Calculating the growth of a hypothetical $1 investment over the analysis
   horizon.
cumulative_returns <- asset_returns %>%
  pivot_longer(-Date, names_to = "Asset", values_to = "Return") %>%
  filter(!is.na(Return)) %>%
  group_by(Asset) %>%
  arrange(Date) %>%
  mutate(Cumulative = exp(cumsum(Return)) / exp(first(Return))) %>%
  ungroup()
```

## A.4 Descriptive Statistical Profiling

```r
# Descriptive Statistics Table
stats_table <- data.frame(
  Mean = apply(asset_returns[active_tickers], 2, mean, na.rm = TRUE),
  Median = apply(asset_returns[active_tickers], 2, median, na.rm = TRUE),
  SD = apply(asset_returns[active_tickers], 2, sd, na.rm = TRUE),
  Min = apply(asset_returns[active_tickers], 2, min, na.rm = TRUE),
  Max = apply(asset_returns[active_tickers], 2, max, na.rm = TRUE),
  Skewness = apply(asset_returns[active_tickers], 2, skewness, na.rm = TRUE),
  Kurtosis = apply(asset_returns[active_tickers], 2, kurtosis, na.rm = TRUE)
)

stats_table
```

## A.5 Multivariate OLS Modeling Workflow

```r
# Robust regression workflow using nest/map
```

```r
model_results <- final_data_long %>%
  group_by(Asset) %>%
  nest() %>%
  mutate(
    model = map(data, ~ lm(Excess_Return ~ Mkt_RF + SMB + HML + RMW + CMA,
   data = .x)),
    tidied = map(model, tidy),
    glanced = map(model, glance)
  ) %>%
  unnest(tidied) %>%
  select(Asset, term, estimate, std.error, statistic, p.value)

# Extract model fitness (Adjusted R-squared)
model_glance <- final_data_long %>%
  group_by(Asset) %>%
  nest() %>%
  mutate(
    model = map(data, ~ lm(Excess_Return ~ Mkt_RF + SMB + HML + RMW + CMA,
   data = .x)),
    glanced = map(model, glance)
  ) %>%
  unnest(glanced) %>%
  select(Asset, r.squared, adj.r.squared, p.value)

# Pivot for summary view
regression_summary <- model_results %>%
  select(Asset, term, estimate, p.value) %>%
  pivot_wider(names_from = term, values_from = c(estimate, p.value))

regression_summary
```

## A.6 Visualization and Custom Theme

```r
# 0. Consistent Color Palette & Custom Theme
asset_colors <- c(
  "VWRA.L" = "#2C3E50",  # Benchmark (Dark Navy)
  "VDEV.L" = "#5D6D7E",  # Developed
  "WSML.L" = "#8E44AD",  # Small Cap
  "QUAL"   = "#27AE60",  # Quality
  "VIG"    = "#3498DB",  # Dividend
  "XLK"    = "#E67E22",  # Tech
  "IAU"    = "#F1C40F",  # Gold
  "VUSD.L" = "#C0392B"   # S&P 500 (Red)
)
```

```r
theme_quant <- function() {
  theme_minimal() +
    theme(
      plot.title = element_text(face = "bold", size = 14, margin = margin(b =
    5)),
      plot.subtitle = element_text(size = 11, color = "#555555", margin =
    margin(b = 15)),
      plot.caption = element_text(size = 8, color = "#777777", hjust = 0,
    margin = margin(t = 15)),
      axis.title = element_text(size = 10, face = "bold"),
      axis.text = element_text(size = 9),
      panel.grid.minor = element_blank(),
      panel.grid.major = element_line(color = "#EBEBEB", linewidth = 0.3),
      legend.position = "bottom",
      legend.title = element_text(size = 10, face = "bold"),
      legend.text = element_text(size = 9),
      axis.ticks = element_blank(),
      plot.margin = margin(20, 20, 20, 20)
    )
}

# 1. Coefficient Bar Chart (Factor Loadings)
coef_plot <- model_results %>%
  filter(term != "(Intercept)") %>%
  ggplot(aes(x = term, y = estimate, fill = Asset)) +
  geom_bar(stat = "identity", position = "dodge", alpha = 0.9) +
  theme_quant() +
  scale_fill_manual(values = asset_colors) +
  labs(title = "Asset Sensitivities to Fama-French Risk Factors",
       subtitle = "Beta Coefficients (Loadings) estimated via OLS Regression",
       x = "Fama-French Factor",
       y = "Beta Loading (Unitless)",
       fill = "Asset Ticker",
       caption = "Note: Factors include Mkt_RF, SMB, HML, RMW, and CMA.\
    nSource: Kenneth French Data Library (2015-2026).") +
  geom_hline(yintercept = 0, color = "black", linewidth = 0.5)

# 2. Model Accuracy (Adjusted R-squared)
rsquared_plot <- model_glance %>%
  ggplot(aes(x = reorder(Asset, adj.r.squared), y = adj.r.squared, fill =
    Asset)) +
  geom_bar(stat = "identity", width = 0.7) +
  coord_flip() +
  theme_quant() +
  scale_fill_manual(values = asset_colors) +
  scale_y_continuous(limits = c(0, 1), breaks = seq(0, 1, 0.2)) +
  labs(title = "Model Explanatory Power: Adjusted R-squared",
```

```r
       subtitle = "Proportion of variance explained by the 5-factor model",
       x = NULL,
       y = "Adjusted R-squared Value",
       fill = "Asset",
       caption = "Higher values indicate better model fit.")

# 3. Cumulative Returns Plot
cum_plot <- cumulative_returns %>%
  ggplot(aes(x = Date, y = Cumulative, color = Asset)) +
  geom_line(linewidth = 0.8) +
  theme_quant() +
  scale_y_log10(breaks = c(1, 1.5, 2, 3, 4, 5)) +
  scale_color_manual(values = asset_colors) +
  labs(title = "Wealth Index: Cumulative Growth of $1 Investment",
       subtitle = "Historical Performance (Logarithmic Scale), 2015 - 2026",
       x = "Year",
       y = "Value of $1 (USD)",
       caption = "Source: Yahoo Finance. First date normalized to 1.00.")

# 4. Security Market Line (SML) Scatter Plot
scatter_data <- final_data_long %>%
  group_by(Asset) %>%
  summarise(
    Mean_Excess = mean(Excess_Return, na.rm = TRUE),
    SD = sd(Return, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  left_join(model_results %>% filter(term == "Mkt_RF") %>% select(Asset,
   estimate), by = "Asset")

scatter_plot <- scatter_data %>%
  mutate(Mean_Excess_Pct = Mean_Excess * 100) %>%
  ggplot(aes(x = estimate, y = Mean_Excess_Pct, label = Asset)) +
  geom_smooth(method = "lm", se = FALSE, color = "#CCCCCC", linetype = "dashed
   ", linewidth = 0.5) +
  geom_point(aes(size = SD, color = Asset), alpha = 0.8) +
  ggrepel::geom_text_repel(size = 3, fontface = "bold", box.padding = 0.5) +
  theme_quant() +
  scale_color_manual(values = asset_colors) +
  scale_size_area(max_size = 10) +
  labs(title = "Security Market Line (SML): Risk-Adjusted Returns",
       subtitle = "Mean Monthly Excess Return (%) vs. Market Beta (Loading)",
       x = "Market Beta (beta)",
       y = "Mean Monthly Excess Return (%)",
       size = "Volatility (SD)",
       caption = "Dashed line represents the empirical SML.")
```

```r
# 5. Rolling Beta Analysis (Stability Diagnostic)
rolling_beta_data <- final_data %>%
  select(Date, VUSD.L, Mkt_RF, RF) %>%
  mutate(VUSD_Excess = VUSD.L - RF)

roll_reg <- function(d) {
  coef(lm(VUSD_Excess ~ Mkt_RF, data = as.data.frame(d)))[2]
}

rolling_beta_data$Beta <- zoo::rollapply(rolling_beta_data %>% select(VUSD_
    Excess, Mkt_RF),
                                          width = 24, FUN = roll_reg, by.column
     = FALSE, fill = NA)

rolling_beta_plot <- rolling_beta_data %>%
  filter(!is.na(Beta)) %>%
  ggplot(aes(x = Date, y = Beta)) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "#C0392B", linewidth
     = 0.8) +
  geom_line(color = "#C0392B", linewidth = 1) +
  theme_quant() +
  labs(title = "Stability Analysis: 24-Month Rolling Market Beta",
       subtitle = "Asset: VUSD.L (S&P 500 ETF) relative to Market Factor",
       x = "Analysis Period",
       y = "Rolling Beta Loading",
       caption = "Red dashed line at 1.00 represents market-neutral
    sensitivity.")

# Create directories and save plots for Final Paper integration
if (!dir.exists("image/paper")) dir.create("image/paper", recursive = TRUE)
ggsave("image/paper/factor_loadings.png", coef_plot, width = 10, height = 6)
ggsave("image/paper/model_accuracy.png", rsquared_plot, width = 8, height = 5)
ggsave("image/paper/cumulative_returns.png", cum_plot, width = 10, height = 6)
ggsave("image/paper/risk_return_scatter.png", scatter_plot, width = 8, height
    = 6)
ggsave("image/paper/rolling_beta.png", rolling_beta_plot, width = 10, height =
     5)

# Display active plots
coef_plot
rsquared_plot
cum_plot
scatter_plot
rolling_beta_plot
```

## A.7 Model Diagnostics and Auditing

```
# 1. Model Fitness Summary (Sorted by Adjusted R-squared)
fitness_summary <- model_glance %>%
  arrange(desc(adj.r.squared)) %>%
  rename(`R-squared` = r.squared, `Adj R-squared` = adj.r.squared, `Model P-
    value` = p.value)

# 2. Significance Audit (Binary Yes/No at 95% Confidence)
significance_summary <- model_results %>%
  mutate(significant = ifelse(p.value < 0.05, "Yes", "No")) %>%
  select(Asset, term, significant) %>%
  pivot_wider(names_from = term, values_from = significant)


# 3. Formatted Coefficient Table (Stars)
formatted_coeffs <- model_results %>%
  mutate(
    stars = case_when(
      p.value < 0.001 ~ "***",
      p.value < 0.01  ~ "**",
      p.value < 0.05  ~ "*",
      TRUE            ~ ""
    ),
    estimate_star = paste0(round(estimate, 4), stars)
  ) %>%
  select(Asset, term, estimate_star) %>%
  pivot_wider(names_from = term, values_from = estimate_star)

# Display Diagnostic Tables
print("--- MODEL FITNESS (ADJUSTED R-SQUARED) ---")
print(fitness_summary)

print("--- FACTOR SIGNIFICANCE AUDIT (P < 0.05) ---")
print(significance_summary)

print("--- REGRESSION COEFFICIENTS WITH SIGNIFICANCE STARS ---")
print(formatted_coeffs)
```